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title: “SALES_DATA” author: “wyne maina” date: “2026-03-04” output: pdf_document —

LOAD THE NECESSARY PACKAGES

```
library("ggplot2")
library("dplyr")
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

SET THE WORKING DIRECTORY

```
setwd("C:/Users/Admin/Desktop/DML")
```

LOAD THE WORKING DIRECTORY

```
library(readr)
df <- read_csv("sales_data (1).csv")
```

```
## Rows: 20 Columns: 5
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): ProductCategory, PaymentMethod, Branch
## dbl (2): OrderID, Quantity
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
print(df)
```

```
## # A tibble: 20 x 5
##    OrderID ProductCategory PaymentMethod Branch  Quantity
##      <dbl> <chr>           <chr>         <chr>      <dbl>
##  1    1001 Beverages       Cash          Nairobi       10
##  2    1002 Food            Card          Mombasa        5
```

```
##  3    1003 Electronics     Cash          Nairobi        2
##  4    1004 Beverages       Card          Kisumu         8
##  5    1005 Food            Cash          Nairobi        6
##  6    1006 Electronics     Card          Mombasa        1
##  7    1007 Beverages       Cash          Kisumu        12
##  8    1008 Food            Card          Nairobi        4
##  9    1009 Electronics     Cash          Mombasa        3
## 10    1010 Beverages       Card          Nairobi        7
## 11    1011 Food            Cash          Kisumu         9
## 12    1012 Electronics     Card          Nairobi        2
## 13    1013 Beverages       Cash          Mombasa        5
## 14    1014 Food            Card          Kisumu         6
## 15    1015 Electronics     Cash          Nairobi        4
## 16    1016 Beverages       Card          Mombasa        3
## 17    1017 Food            Cash          Nairobi        8
## 18    1018 Electronics     Card          Kisumu         2
## 19    1019 Beverages       Cash          Nairobi        9
## 20    1020 Food            Card          Mombasa        7
```

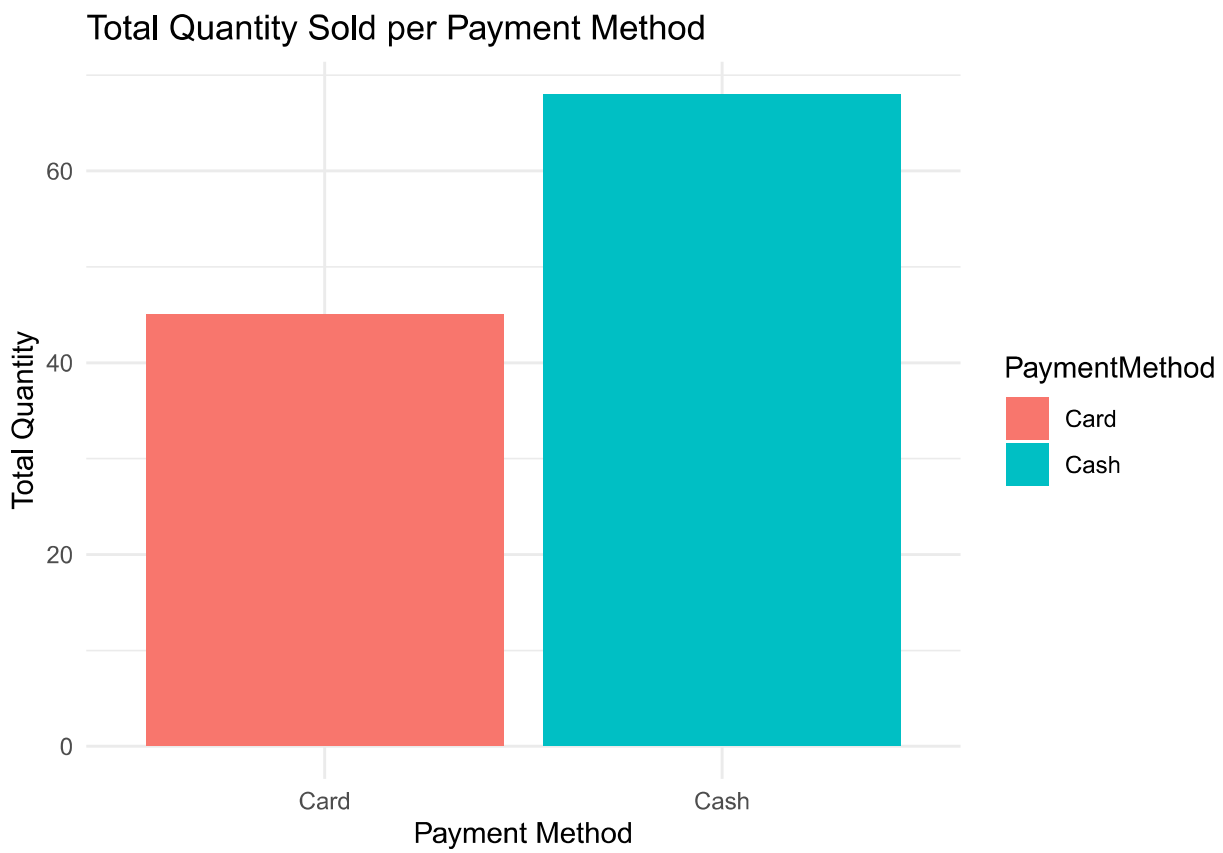
column naming

```r
colnames(df)
```

```
## [1] "OrderID"         "ProductCategory" "PaymentMethod"   "Branch"
## [5] "Quantity"
```

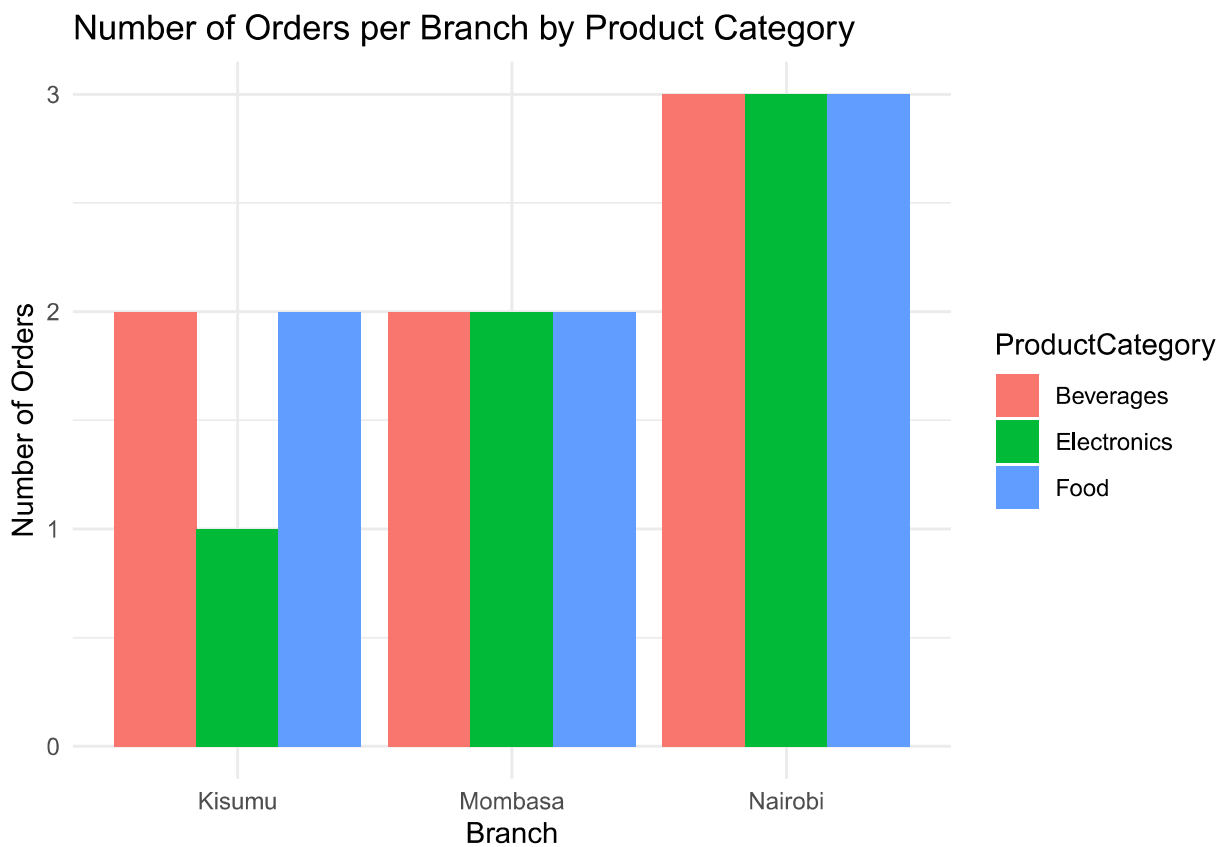
#1. Total Quantity Sold per PaymentMethod

```r
ggplot(df, aes(x = PaymentMethod, y = Quantity, fill = PaymentMethod)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Total Quantity Sold per Payment Method",
    x = "Payment Method",
    y = "Total Quantity"
  ) +
  theme_minimal()
```



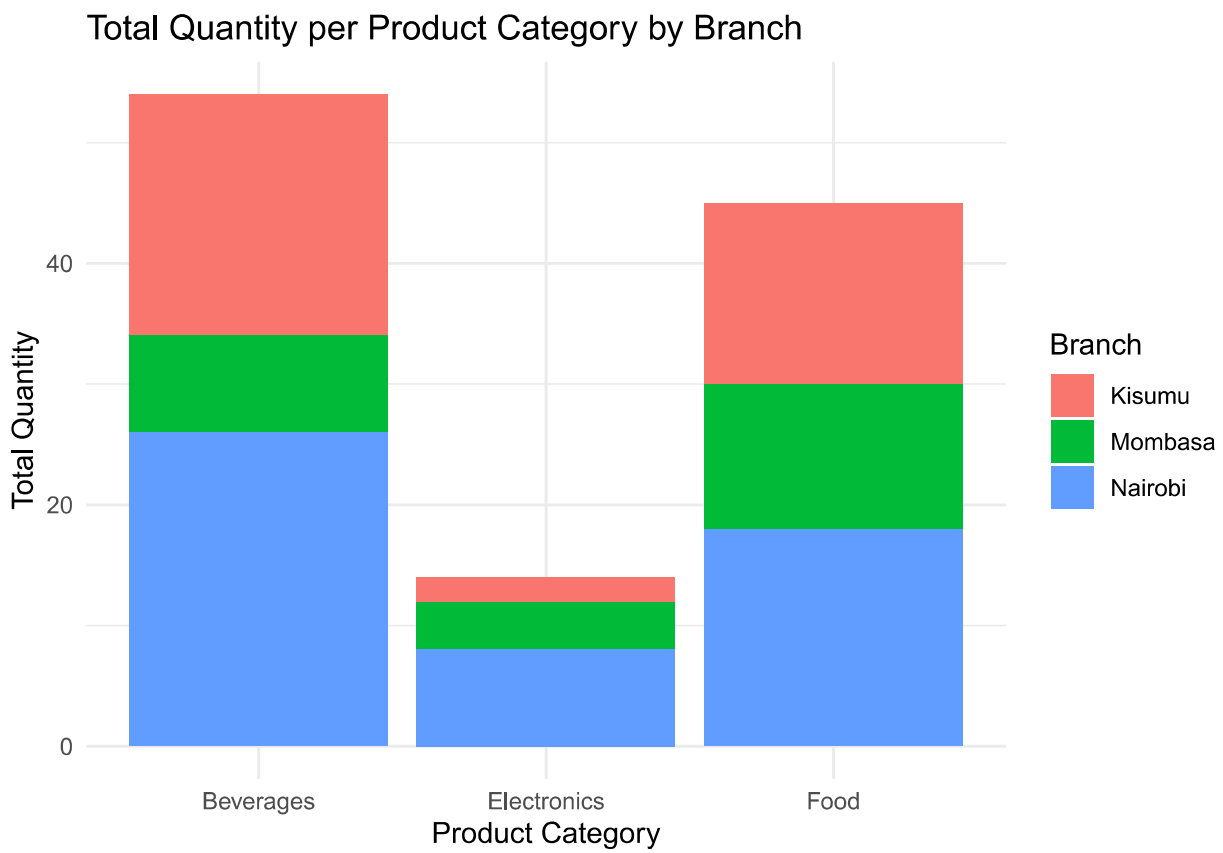
##2. Grouped Bar Chart: Number of Orders per Branch by ProductCategory

```
ggplot(df, aes(x = Branch, fill = ProductCategory)) +
  geom_bar(position = "dodge") +
  labs(
    title = "Number of Orders per Branch by Product Category",
    x = "Branch",
    y = "Number of Orders"
  ) +
  theme_minimal()
```



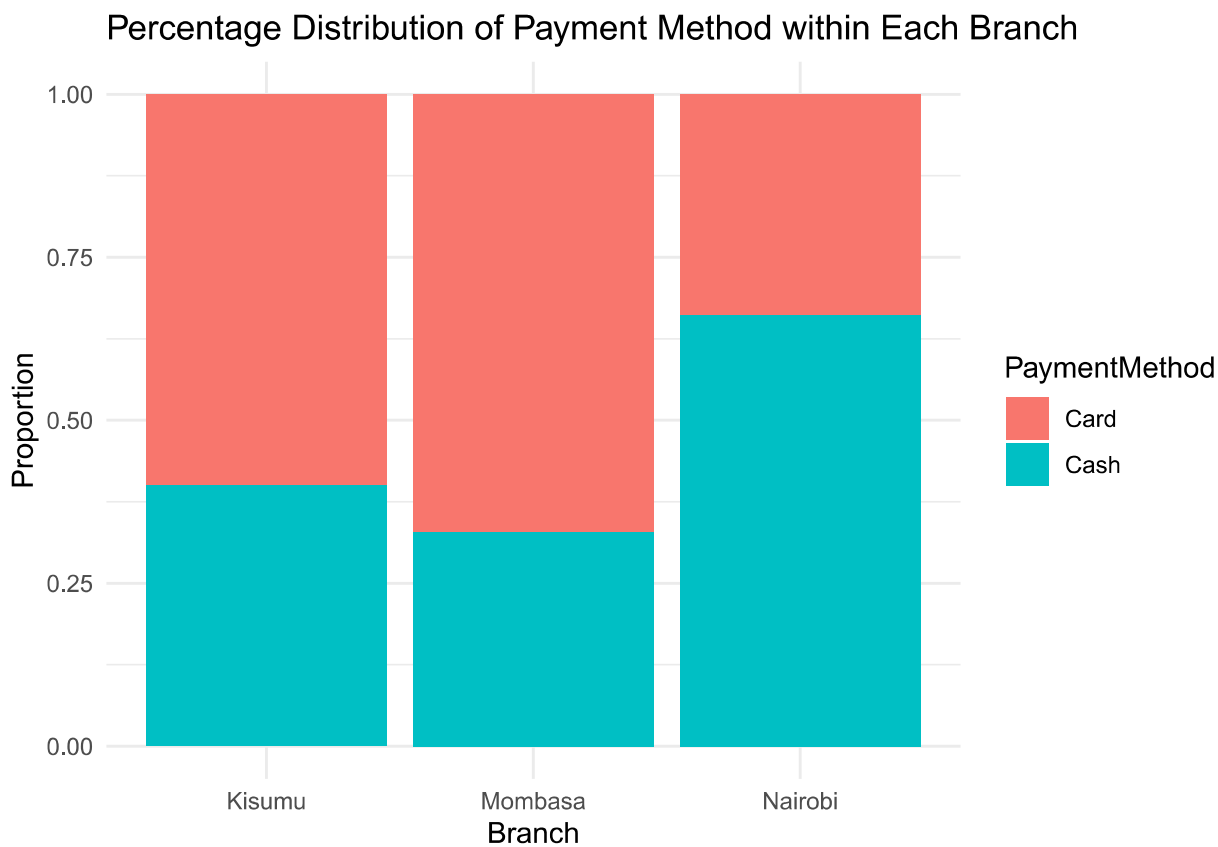
#3. Stacked Bar Chart: Total Quantity per ProductCategory split by Branch

```
ggplot(df, aes(x = ProductCategory, y = Quantity, fill = Branch)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Total Quantity per Product Category by Branch",
    x = "Product Category",
    y = "Total Quantity"
  ) +
  theme_minimal()
```



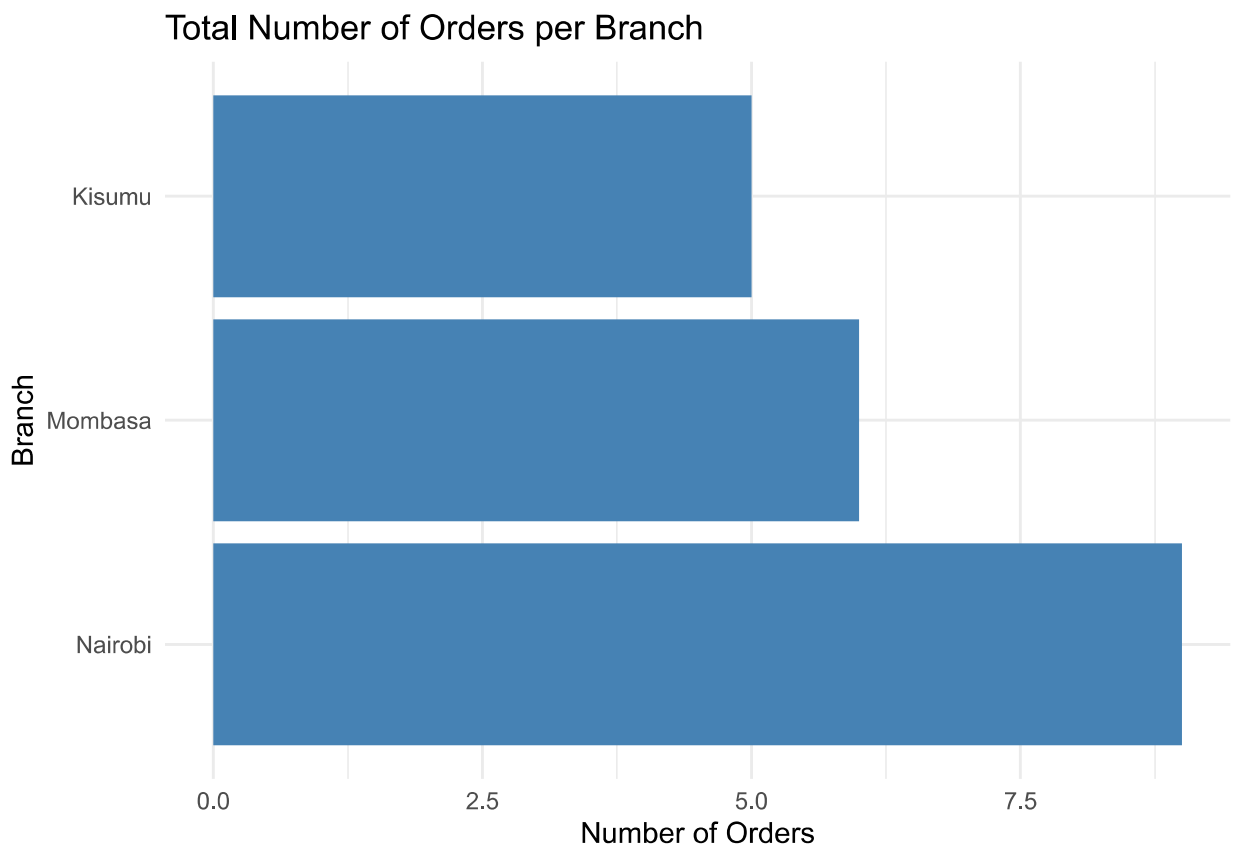
##4. Proportional Bar Chart: Percentage Distribution of PaymentMethod within each Branch

```
ggplot(df, aes(x = Branch, fill = PaymentMethod)) +
  geom_bar(position = "fill") +
  labs(
    title = "Percentage Distribution of Payment Method within Each Branch",
    x = "Branch",
    y = "Proportion"
  ) +
  theme_minimal()
```



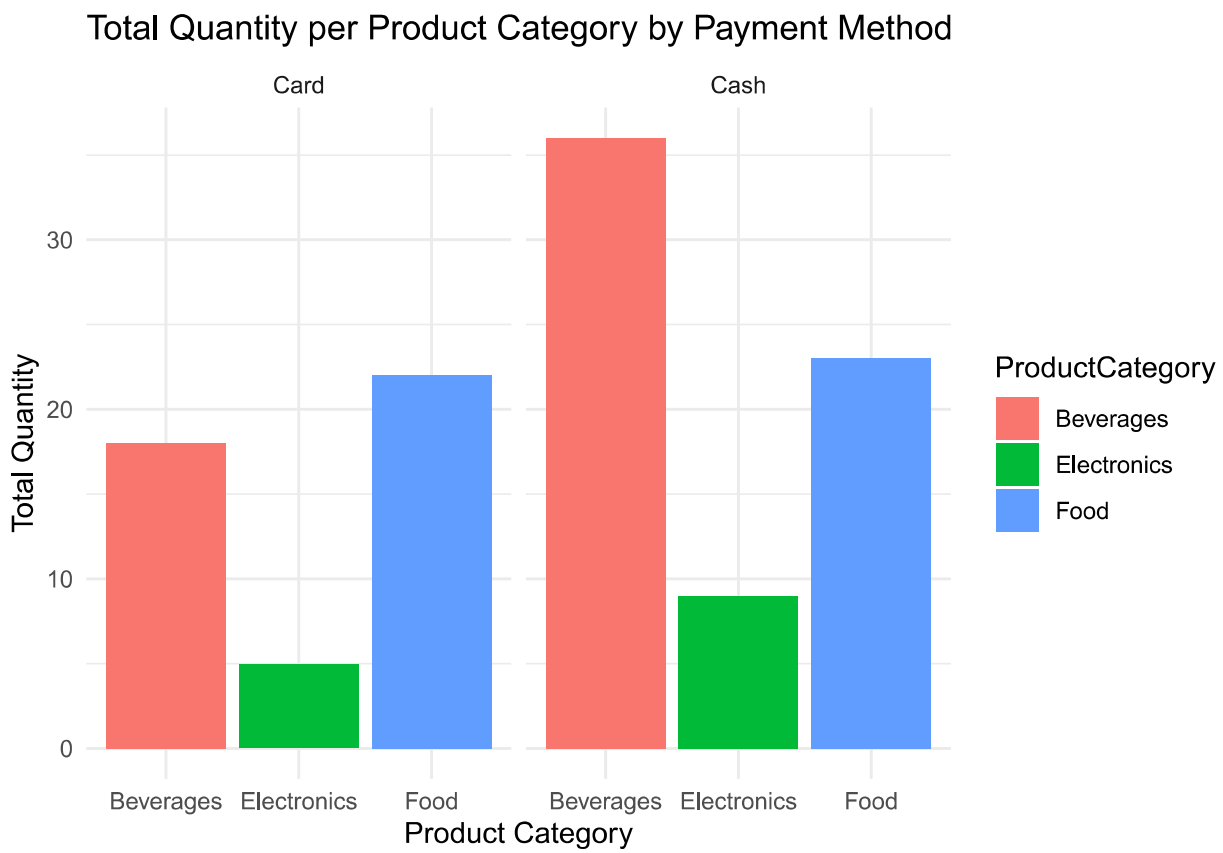
#5. Horizontal Bar Chart: Total Number of Orders per Branch (sorted)

```
ggplot(df, aes(x=
                 forcats::fct_infreq(Branch))) +
  geom_bar(fill = "steelblue") +
  coord_flip() +
  labs(
    title = "Total Number of Orders per Branch",
    x = "Branch",
    y = "Number of Orders"
  ) +
  theme_minimal()
```



##6 Faceted Bar Chart: Total Quantity per ProductCategory, separate panels for each PaymentMethod

```
ggplot(df, aes(x = ProductCategory, y = Quantity, fill = ProductCategory)) +
  geom_bar(stat = "identity") +
  facet_wrap(~ PaymentMethod) +
  labs(
    title = "Total Quantity per Product Category by Payment Method",
    x = "Product Category",
    y = "Total Quantity"
  ) +
  theme_minimal()
```


Total Quantity per Product Category by Payment Method
Card
Cash
Total Quantity
30
20
10
0
Beverages
Electronics
Food
Beverages
Electronics
Food
Product Category
ProductCategory
Beverages
Electronics
Food

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